

Total No. of Questions : 8]

[Total No. of Printed Pages : 3

Roll No

MI-5004-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Mechanics of Solids and Fluids

Time : Three Hours

Maximum Marks : 70

- Note :** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain the terms: 7
i) Strain
ii) Shear strain
iii) Young's modulus
iv) Modulus of rigidity
b) What do you mean by Temperature stresses? Explain. 7
2. a) How are beams classified? Give a brief account. 7
b) A simply supported beam of 8m span carries point loads of 24kN and 40kN at distances of 2m from each end. Draw the shear force and bending moment diagrams. 7
3. a) Establish the governing differential equation of beams. What are its limitations? 7
b) A cantilever of 3m span carries a uniformly distributed load of 10kN per meter length for 2m starting from the fixed end. Determine the slope and the deflection of the free end. $E = 200\text{Gpa}$ and $I = 80 \times 10^6\text{m}^4$. 7

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[2]

4. a) Briefly explain the significance of fluid mechanics and the vital role it play in a variety of engineering applications. 7
- b) Distinguish between: 7
 - i) Ideal and real fluids
 - ii) Compressible and incompressible fluids
5. a) Explain the terms center of buoyancy, metacentre and metacentric height. 7
- b) Explain the term total pressure acting on a plane surface immersed in a fluid at any angle. Obtain an expression for this and also for the corresponding depth of the centre of pressure. 7
6. a) What do you mean by principal planes and principal stresses? Derive the expression for principal stresses for a body subjected to direct and shear stresses. 7
- b) Two rectangular bars, one of brass and the other of steel, each 36mm by 9mm are placed together to form a beam of 36mm by 18mm depth on two supports 800mm apart, the steel bar being at the bottom. Find the maximum central load if the bars are 7
 - i) Separate and can bend independently
 - ii) Firmly secure to each other throughout the lengthMaximum permissible stress in steel is 102Mpa and in brass 72Mpa. Take $E_s = 204 \text{ Mpa}$ and $E_b = 85 \text{ Mpa}$.
7. a) What is Macaulay's method of beam deflection analysis? What are its advantages over the direct integration method? 7
- b) A diver is working at a depth of 20m below the surface of sea water (sp.wt. = 10 kN/m^3). Calculate the pressure intensity at this depth. What would be the absolute pressure if barometer reads 760mm of mercury column at the sea level? 7

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[3]

8. a) Define : 7
- i) Stream function
 - ii) Potential function
- b) Classify flow: 7
- i) Uniform and Non-Uniform flow
 - ii) Steady and Non-Steady flow
 - iii) Laminar and Turbulent flow
